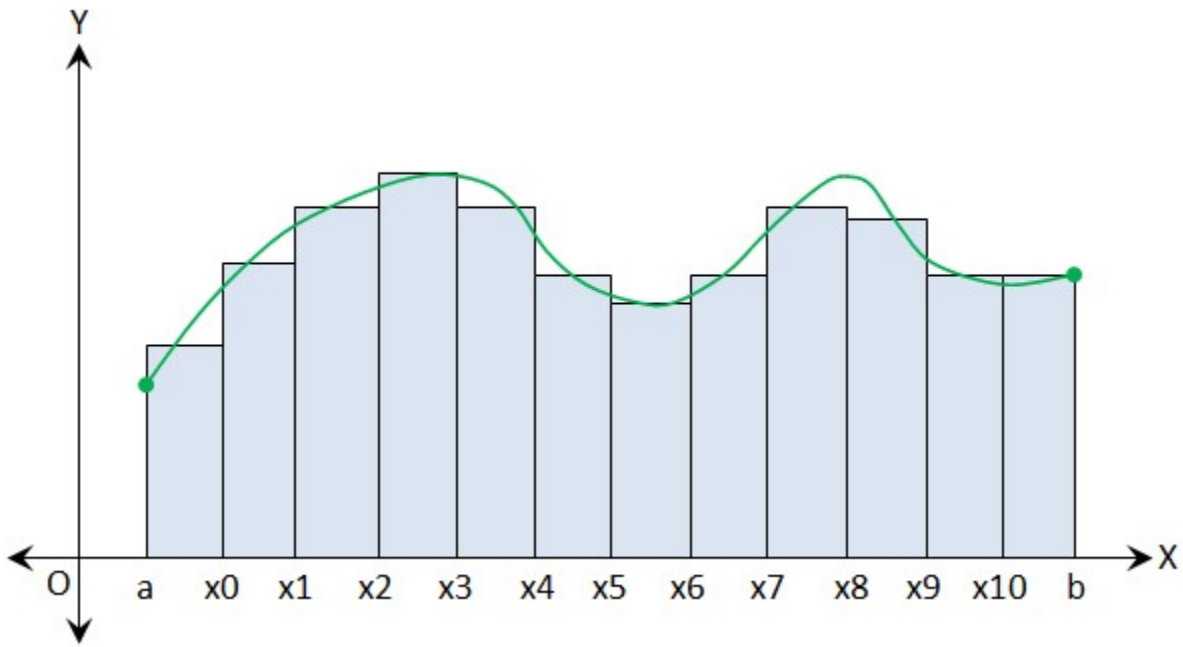
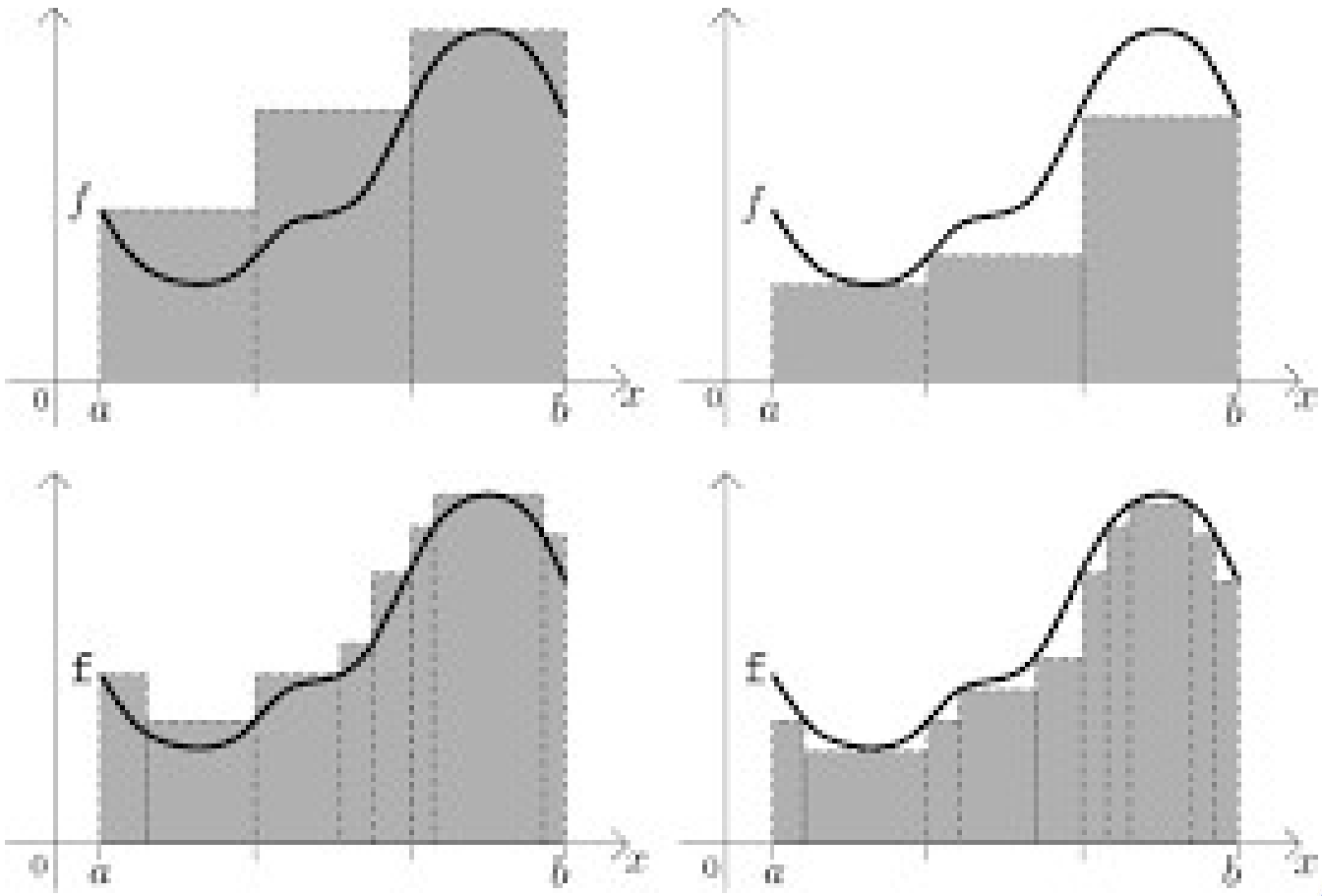
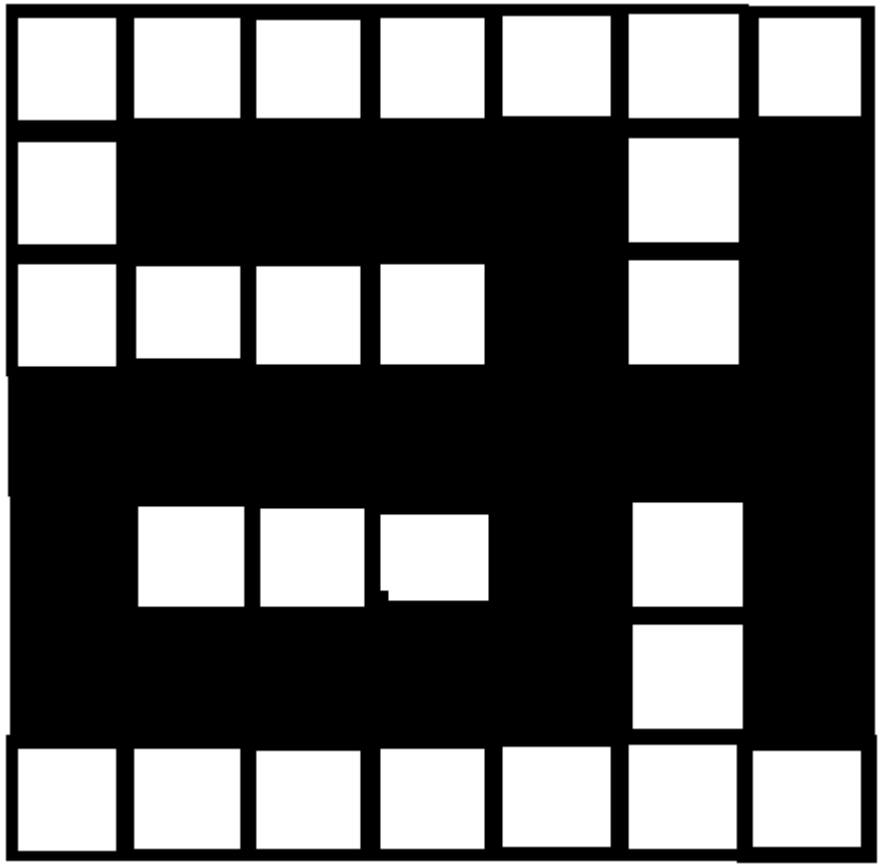
Digital Image Representation

Approximating the curve



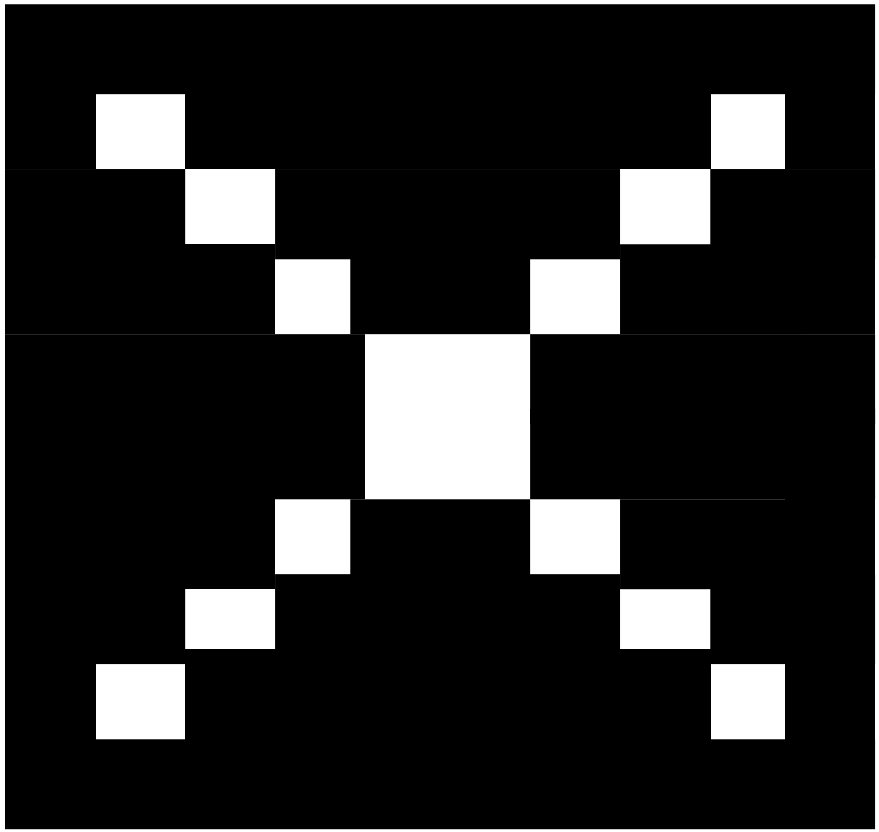
Coarse and fine grids

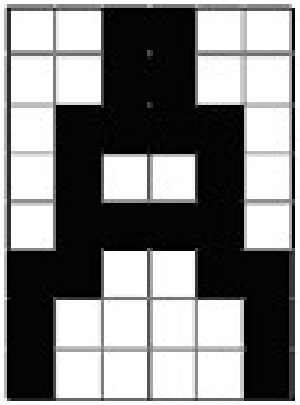




Black and white image

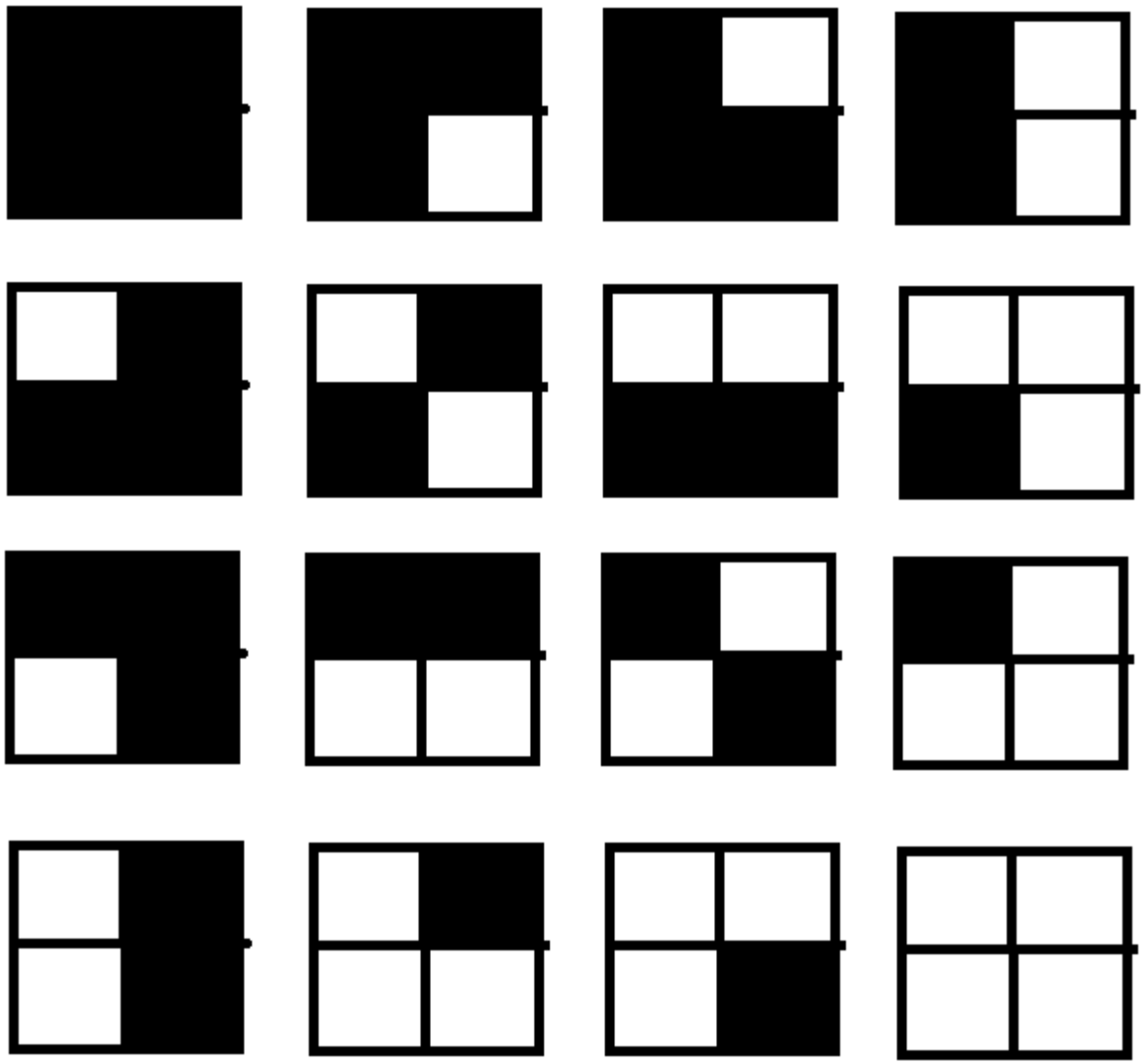
0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	1	0
0	0	1	0	0	0	0	1	0	0
0	0	0	1	0	0	1	0	0	0
0	0	0	0	1	1	0	0	0	0
0	0	0	0	1	1	0	0	0	0
0	0	0	1	0	0	1	0	0	0
0	0	1	0	0	0	0	1	0	0
0	1	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0	0





0	0	1	1	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	1	0	0	1	0
0	1	1	1	1	0
1	1	0	0	1	1
1	0	0	0	0	1
1	0	0	0	0	1

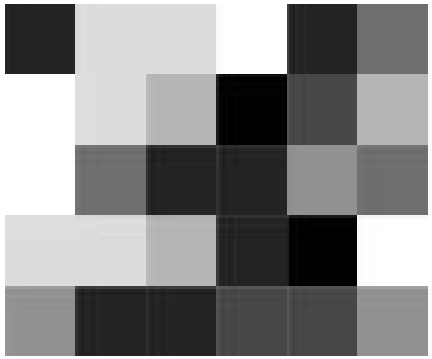
Assume (2x2) pixel camera



Possible images

Why only black and white?

- We use in between colours of black and white
- Various gray levels
- 6 – black
- 0 – white
- {1, 2, 3, 4, 5} - various gray levels



$$\Longleftrightarrow \begin{bmatrix} 6 & 1 & 1 & 0 & 6 & 4 \\ 0 & 1 & 2 & 7 & 5 & 2 \\ 0 & 4 & 6 & 6 & 3 & 4 \\ 1 & 1 & 2 & 6 & 7 & 0 \\ 3 & 6 & 6 & 5 & 5 & 3 \end{bmatrix}$$

How many levels?

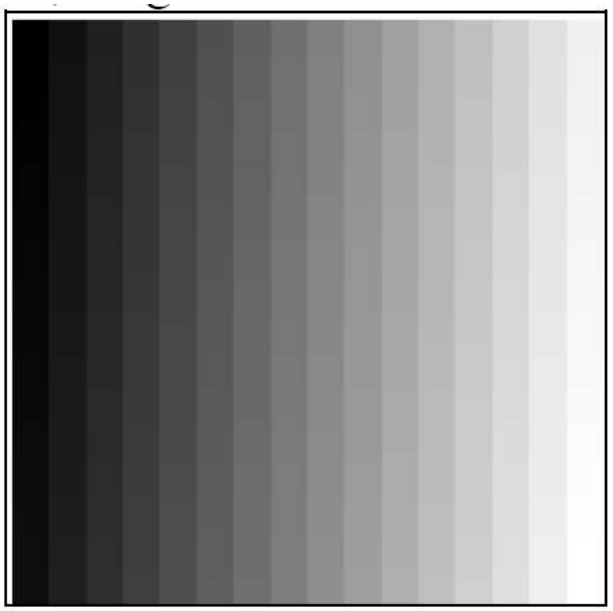
- With 2 levels (i.e. black and white) details are lost
- Introducing some gray levels $\rightarrow$ Better image quality

Gray scale image - Standardization

- 0 – black
- 255 – white
- {1, 2, 3, …, 254} – various gray levels
- Total levels 256
- We need 8 bits to represent 256 gray levels

Colour Image

- R + G + B
- Each represented using 8 bits
- Total bits 24
- How many levels? 2^{24}
- 24 bit representation aka “True colour”



0	16	32	48	64	80	96	112	128	144	160	176	192	208	224	240
1	17	33	49	65	81	97	113	129	145	161	177	193	209	225	241
2	18	34	50	66	82	98	114	130	146	162	178	194	210	226	242
3	19	35	51	67	83	99	115	131	147	163	179	195	211	227	243
4	20	36	52	68	84	100	116	132	148	164	180	196	212	228	244
5	21	37	53	69	85	101	117	133	149	165	181	197	213	229	245
6	22	38	54	70	86	102	118	134	150	166	182	198	214	230	246
7	23	39	55	71	87	103	119	135	151	167	183	199	215	231	247
8	24	40	56	72	88	104	120	136	152	168	184	200	216	232	248
9	25	41	57	73	89	105	121	137	153	169	185	201	217	233	249
10	26	42	58	74	90	106	122	138	154	170	186	202	218	234	250
11	27	43	59	75	91	107	123	139	155	171	187	203	219	235	251
12	28	44	60	76	92	108	124	140	156	172	188	204	220	236	252
13	29	45	61	77	93	109	125	141	157	173	189	205	221	237	253
14	30	46	62	78	94	110	126	142	158	174	190	206	222	238	254
15	31	47	63	79	95	111	127	143	159	175	191	207	223	239	255

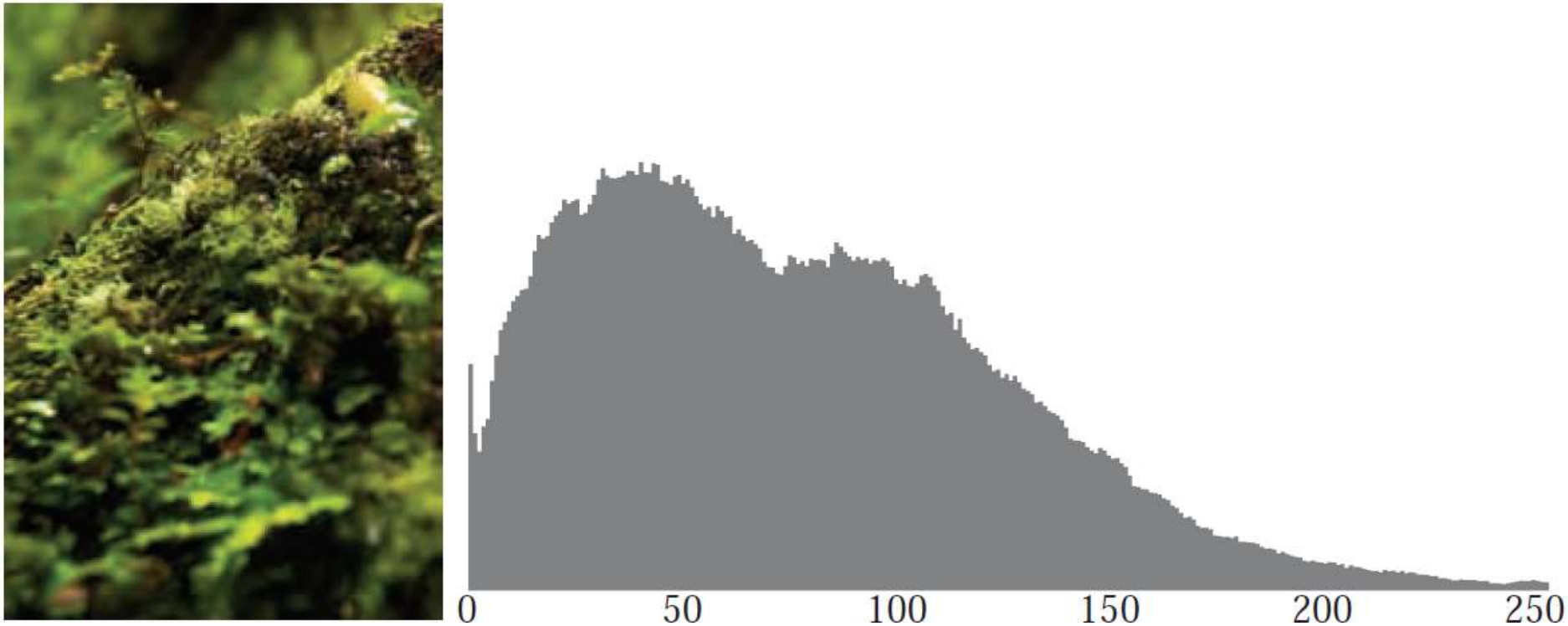
Digital Image

- A digital image ≡ pixels matrix
- An area divided using grid
- Pixel – the smallest area

Histogram of an image

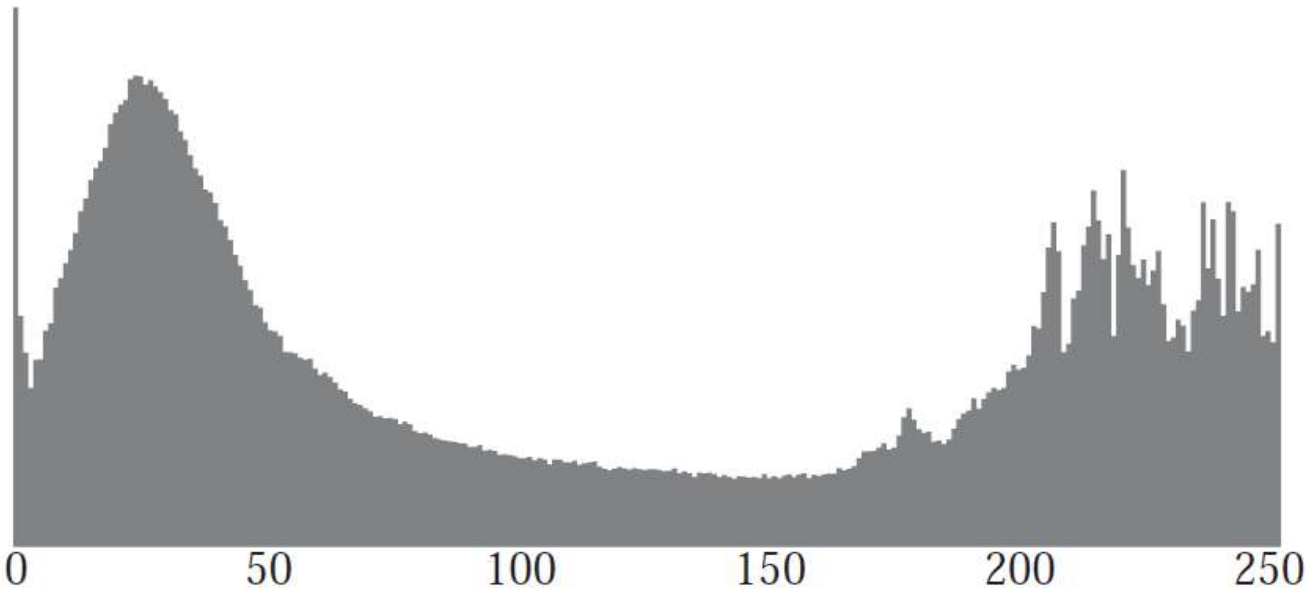
- Distribution of pixel values in images
- Image $\equiv$ matrix with entries {0, 1, 2, …255}
- Count "How many 0s? & How many 1s? & How many 2s? & ……& How many 255 s?"

e.g.1

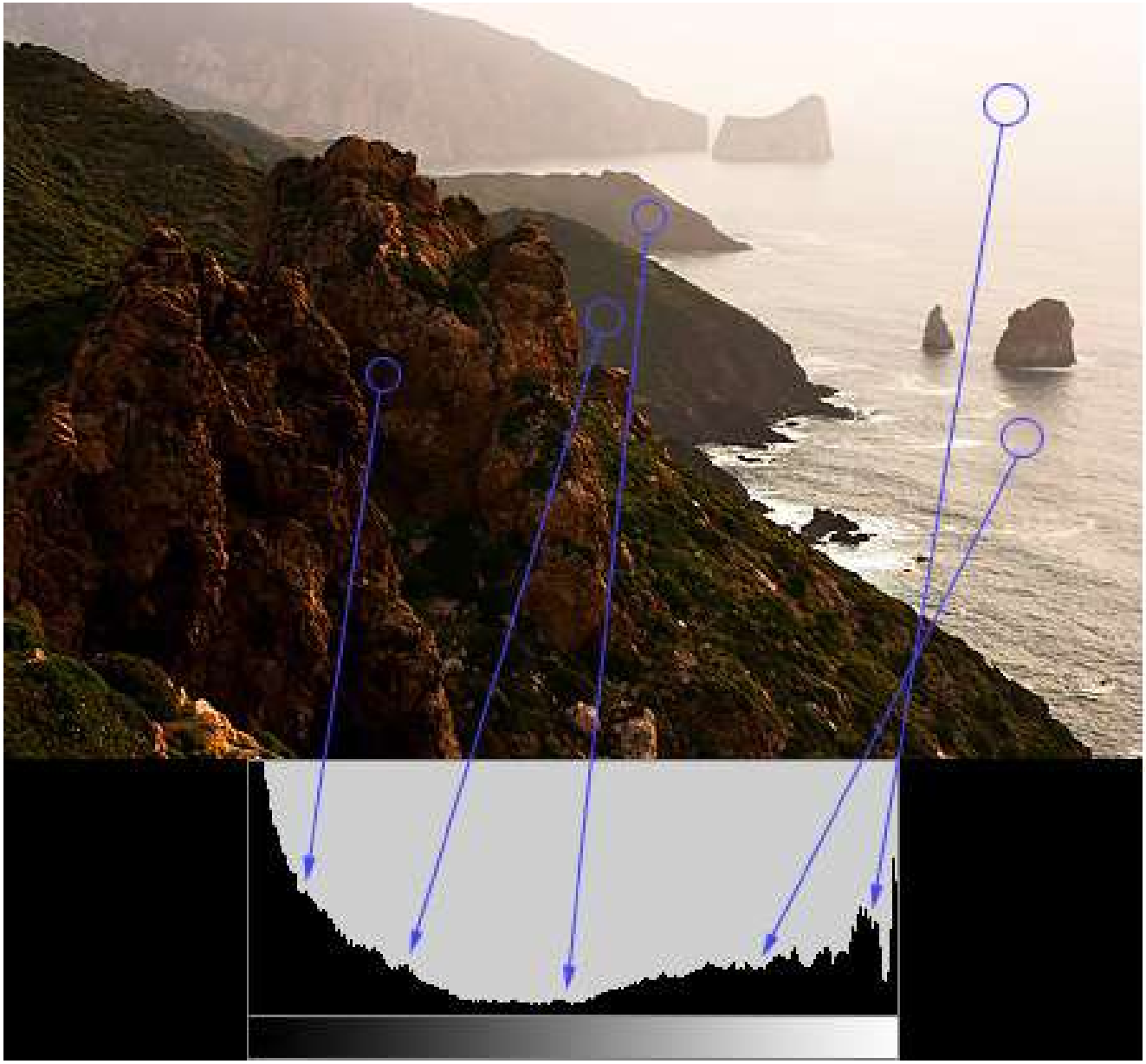


- Image is uniformly illuminated
- Approximately Gaussian distribution

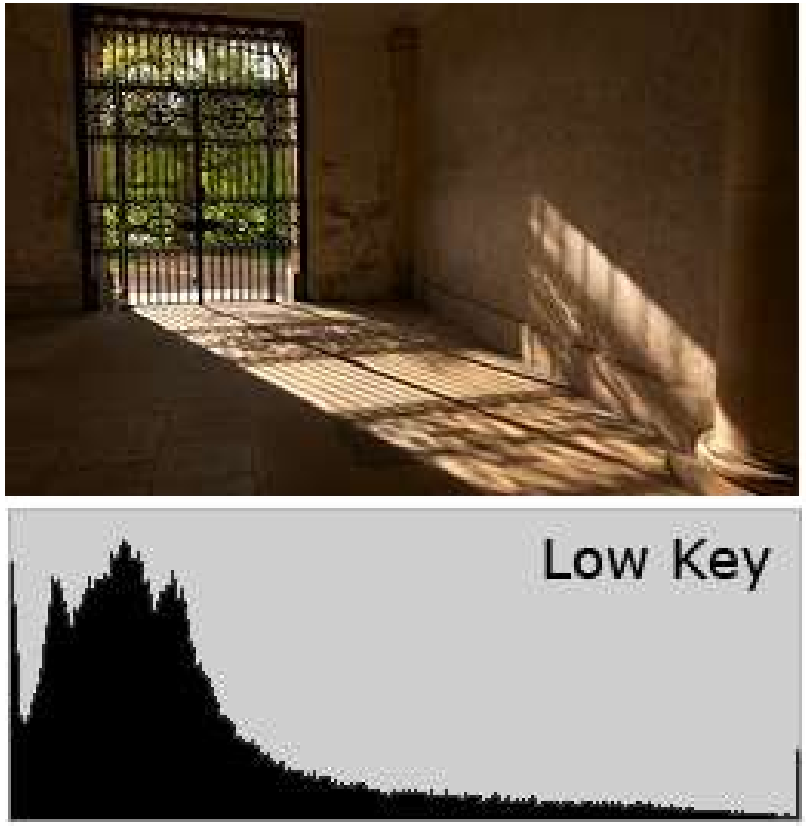
e.g.2

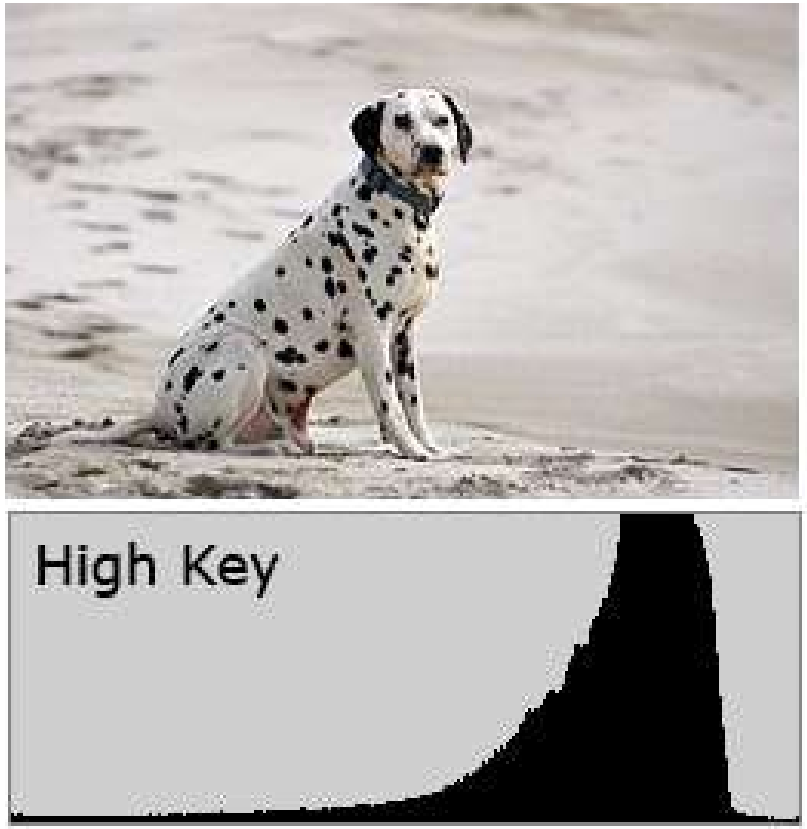


- Parts of the scene are in the shade & other parts are brightly illuminated
- Bimodal distribution



e.g. 3



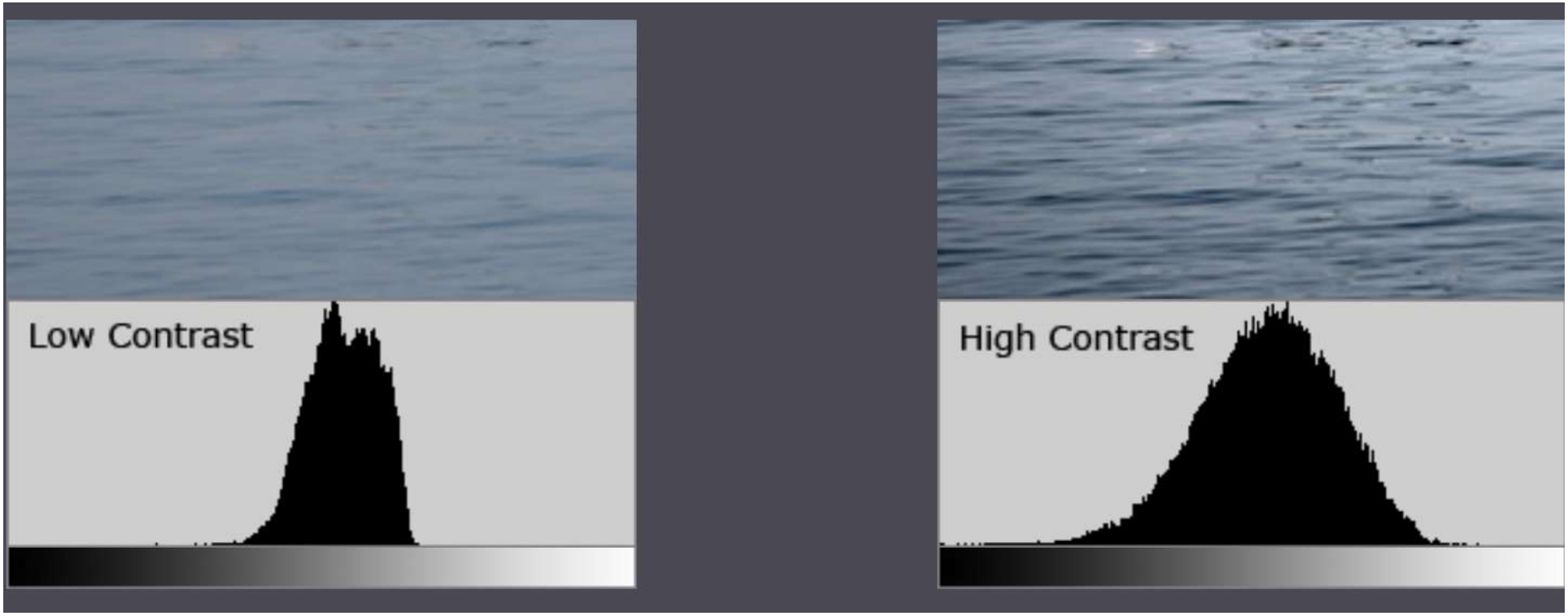


Till now we saw
“Histograms convey brightness information”

What about contrast?

- Brightness: Zeroth order derivative
- Contrast: Kind of first derivative
- Difference in brightness between light and dark regions of the given scene

Lighter and darker regions in histograms

- Darker regions: towards 0
- Lighter regions: towards 255

Wider the histogram greater the contrast